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ALL INDIA INTEGRATED TEST SERIES

FIITJEE

JEE(Main)-2016

ANSWERS, HINTS & SOLUTIONS

HALF COURSE TEST-VI
(Main)

Q. No.	PHYSICS	CHEMISTRY	MATHEMATICS
1.	A	B	A
2.	B	A	A
3.	B	D	B
4.	B	D	A
5.	B	B	D
6.	C	B	A
7.	A	D	B
8.	D	D	B
9.	C	B	B
10.	B	B	B
11.	D	B	B
12.	B	D	D
13.	B	B	B
14.	B	D	C
15.	A	B	A
16.	C	B	C
17.	C	B	A
18.	D	B	B
19.	D	D	D
20.	B	B	C
21.	D	C	B
22.	C	C	A
23.	C	C	B
24.	C	A	B
25.	A	A	A
26.	C	A	A
27.	C	C	B
28.	C	A	D
29.	C	D	B
30.	B	C	A

Physics

PART – I

SECTION – A

1. Friction force on block A, $f_K = \mu N = 10$ N (opposite to v)

$\Rightarrow$ acceleration of A = $\frac{10N}{4kg} = 2.5$ m/s^2 (opposite to v)

Time taken by the plank to come at rest,

$t = \frac{10\ m/s}{2.5\ m/s^2} = 4$ sec (using, v = u + at)

In 4s distance moved by the plank $S_A = 20$ m (using $v^2 = u^2 + 2as$).

To reach the other end block will have to move by 20 m + 40 m = 60 m. As B is moving with constant speed

$\Rightarrow \quad T = \frac{60\ m/s}{10\ m/s} = 6\,sec$

2. Reference frame is at the edge of the plane with the leftward direction as the x-axis. Let x_0 be the distance of the cart at t = 0

mg – T = ma

T = Ma

$\therefore \quad a = \frac{mg}{M+m} = 2.8$ m/s^2

Since, M has an initial velocity to the left, a is in fact retardation for M.

$\therefore \quad V = 7 + (-2.8) \times 5 = -7$ m/s

i.e. after 5 sec the cart will have the same speed but in opposite direction.

$x = x_0 + 7 \times 5 + \frac{1}{2}(-2.8) \times 5$

$x = x_0 + 35 - 35 = x_0$

Let the cart go by S to the left from its initial position

$0^2 - 7^2 = 2(-2.8) \times S$

$\Rightarrow \quad S = 8.75$ m

$\therefore \quad$ Distance covered in up and down journey = 8.75 + 8.75 = 17.5 m

3. Let the pressures be P_1 and P_2 of the spheres having initial temperatures 0°C and 20°C respectively then.

Initially $P_1 = P_2 \Rightarrow \frac{n_1}{n_2} = \frac{293}{273}$

If P_1', P_2' be the pressures after rasing temperature then $\frac{P_1'}{P_2'} = \frac{283}{303} \times \frac{293}{273} > 1 \Rightarrow P_1' > P_2'$

thus the mercury pellet will be displaced towards the sphere at higher temp.

4. No eddy current form in case (i) & (ii) and hence motion is not opposed.

7. $(1.2 \times 10^3 \times 10 \times 10 \times 10^{-2}) \times (10^3 \times 10^{-4}) - F$

= Buoyant force – weight

$\Rightarrow$ F = 20 N in down ward direction

8. Kinetic energy is maximum at centre.

10. $dU = \frac{1}{2}\left(\frac{mgx}{LA}\right)^2 \times \frac{1}{y} \times A.dx$

$$U = \frac{1}{2}\frac{m^2g^2A}{L^2A^2Y}\int_0^l x^2dx$$

$$\Rightarrow U = \frac{m^2g^2l}{6AY}.$$

12. A node is a point where there is no vibration. There is vibration at other points where nodes are not formed.

13. $W = \oint \vec{F}.d\vec{r} = \oint \frac{b\, d\vec{r}}{|d\vec{r}|R}.d\vec{r} = \frac{b}{R}\oint |d\vec{r}| = \frac{2\pi Rb}{R} = 2\pi bJ.$

14. Using $A_1v_1 = A_2v_2$ ……(i)
and Bernoulli's principle we get

$$P_1 + \frac{1}{2}\rho v_1^2 = P_2 + \frac{1}{2}\rho v_2^2 \quad \text{.....(ii)}$$

From (i) and (ii)

$$v_1 = A_2\sqrt{\frac{2(P_1 - P_2)}{\rho(A_1^2 - A_2^2)}}$$

Rate of flow = $A_1v_1 = A_1A_2\sqrt{\frac{2(P_1 - P_2)}{\rho(A_1^2 - A_2^2)}}$

But $P_1 - P_2 = \Delta P$

$\therefore$ rate of flow of glycerine $= A_1A_2\sqrt{\frac{2\Delta P}{\rho(A_1^2 - A_2^2)}}$

15. In adiabatic compression, the temperature always increases and since PV = nRT, the quantity PV also increases.

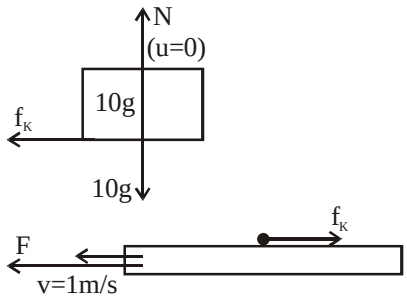
16. $V = 0 + (0.4 \times 10)\, t = 1$ m/s.

$t = \frac{1}{4} = .25$ second.

$W_{f_K} + W_F = \Delta K = 0$

$W_F = -W_{f_K} = -(f_K \times S \times \cos 180°) = f_K \times S$

$= (0.4 \times 10 \times 10) \times .25 \times 1 = 40 \times .25 = 10$ J.



17. At the farthest point X,

$mv_0R = mv_x\, 4R$

$$\frac{1}{2}m v_x^2 - \frac{GMm}{4R} = \frac{1}{2}mv_0^2 - \frac{GMm}{R}$$

$\Rightarrow v_0 = \sqrt{1.6gR}$.

18. T = period $= 2\pi\sqrt{\frac{m}{K_1+K_2}} = 2\pi\sqrt{\frac{10}{360}} = \frac{\pi}{3}s$.

The maximum velocity is always at equilibrium position since at any other point there will be a restoring force attempting to slow the mass.

$\therefore \quad V_{max} = \frac{impulse}{mass} = \frac{50}{10} = 5ms^{-1}$

$A = amplitude = \frac{V_{max}}{\omega} = \frac{5}{\frac{(2\pi)}{\left(\frac{\pi}{3}\right)}} = \frac{5}{6}m.$

19. Their average translational K.E. is $\frac{3}{2}k_B T$, $k_B \rightarrow$ Boltzmann's constant

$\frac{1}{2}m_{H_2}C_{H_2}^2 = \frac{1}{2}m_{He}C_{He}^2 = \frac{3}{2}k_BT$

$C_{H_2} : C_{He} = \sqrt{m_{He}} : \sqrt{m_{H_2}} = \sqrt{4} : \sqrt{2} = 2 : \sqrt{2}$

20. For the mixture, $\gamma = \frac{C_P^{(1)} + C_P^{(2)}}{C_V^{(1)} + C_V^{(2)}} = \frac{\frac{7}{2}R + \frac{5}{2}R}{\frac{5}{2}R + \frac{3}{2}R} = \frac{3}{2}$

Applying $TV^{\gamma - 1}$ = constant, the temperature increases by a factor of $\sqrt{8}$.

21. Impulse = Force $\times$ time; net momentum of the system is conserved as the net force on it is zero.

22. Since velocity of man w.r.t trolley is greater than velocity of trolley w.r.t. earth, after the man turns back displacement of the man will decrease, so maximum displacement will be at the moment when man turns
back.

$\therefore \quad t = \frac{L}{1.5V}$

$\therefore \quad$ Displacement $= (v + 1.5v)t = 2.5v\frac{L}{1.5v} = \frac{5}{3}L$

23. $pV^{1/4}$ = constant

$\Rightarrow \frac{RT}{V}v^{1/4}$ = constant or $TV^{-3/4}$ = constant

$dT\,V^{-3/4} - \frac{3}{4}dV.V^{-7/4}.T = 0$ or $\frac{dV}{dt} = \frac{4}{3}\frac{V}{T}$

$C = \frac{dQ}{dT} = \frac{dU}{dT} + P\frac{dV}{dT} = C_v + p.\frac{4}{3}\frac{V}{T} = \frac{5}{2}R + \frac{4}{3}R = \frac{23}{6}R.$

24. $v = \frac{\upsilon}{4l_1}$ (for first harmonic)

$v = \frac{3\upsilon}{2l_3}$ (for third harmonic)

$\therefore \quad \frac{l_1}{l_3} = \frac{1}{6}$

25. velocity of upper block when lower block hits a obstacle,

$v = \sqrt{2a.\ell} = \sqrt{2.\frac{F}{2M}\ell} = \sqrt{\frac{F\ell}{M}}$

now after collision retardation of upper block w.r.t. earth,

$a = \frac{\mu Mg}{M} = \mu g$

$\therefore \quad v^2 = u^2 + 2as$

$\therefore \quad 0 = \frac{F\ell}{M} - 2\mu g.\frac{\ell}{2} \Rightarrow \mu = \frac{F}{Mg}$.

26. $w = \frac{1}{2}I\omega^2$

Let x is the distance of CM from A

$I = M.x^2 + 4M(\ell - x)^2$ and if I is minimum, W will be minimum

$\therefore \quad \frac{dI}{dx} = 2Mx + 4M \times 2(\ell - x) \times (-1) = 2\,Mx - 8M\,(\ell - x)$

$\frac{dI}{dx} = 10Mx - 8M\ell$

$\therefore \quad 10Mx - 8M\ell = 0$

$x = \frac{4}{5}\ell$.

27. Total internal energy of system will remain conserve

$nC_{V_1}T_1 + nC_{V_2}T_2 = \left[nC_{V_1} + nC_{V_2}\right]T$

$\left(\frac{3}{2}R\right)300 + \left(\frac{5R}{2}\right)600 = \left[\frac{3R}{2} + \frac{5R}{2}\right]T$

$T = \left(\frac{9+30}{8}\right)100 = \frac{3900}{8}$ = 487K (approx)

$\Delta E = \left(\frac{3}{2}R\right)\left[487 - 300\right] = 187 \times \frac{3}{2} \times R$ = 280.5 R

28. Let initial velocity of the shot be u and P denote the force of resistance of penetration.
In the 1st case plate is fixed

$\therefore \quad 0 - \frac{1}{2}mu^2 = -PS$(COE) …(i)

In the 2nd case plate is free to move. The shot will penetrate up to both acquire a common velocity V (say), there will be no further penetration thus P seases to act

$\frac{1}{2}(M+m)V^2 - \frac{1}{2}mu^2 = -PS'$(COE) …(ii)

and $(M+m)V = mu$(COLM) …(iii)

From equation (iii) and (ii)

$\frac{1}{2}\frac{m^2u^2}{(M+m)} - \frac{1}{2}mu^2 = -PS'$

$\Rightarrow \quad \frac{1}{2}mu^2 \times \frac{M}{(M+m)} = PS'$ …(iv)

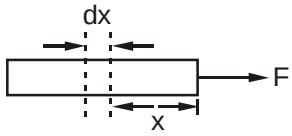
equation (i) and (ii)

$$\frac{S'}{S} = \frac{M}{(M+m)}$$

$$\therefore \quad S' = \frac{SM}{(M+m)}.$$

(COE = conservation of energy)
(COLM = conservation of linear momentum).

29. Stress = $\frac{T}{A}$ E = $\frac{1}{2}$ Stress × Strain × Volume



$$\frac{\text{Stress}}{\text{Strain}} = Y, \quad E = \frac{1}{2}\frac{(\text{Stress})^2}{Y} \times \text{Volume}$$

$$F = ma, \quad E = \int_o^L \frac{1}{2}\left(\frac{L-x}{L}\frac{F}{A}\right)^2 \frac{1}{Y} A\, dx$$

$$T = (L-x)\lambda \quad \frac{F}{M} = \frac{F^2 L}{6YA} = \frac{L-x}{L}F$$

30. $\upsilon_0 - at = 0 \quad \Rightarrow \quad \upsilon = at$

$\omega_0 - \alpha t = 0 \quad \Rightarrow \quad \omega_0 = \alpha t$

$$\Rightarrow \quad \frac{v}{\omega_0} = \frac{a}{\alpha} = \frac{r}{2}$$

$$\Rightarrow \quad \frac{v}{\omega_0 r} = \frac{1}{2}$$

Chemistry

PART – II

SECTION – A

4. $Na \longrightarrow 1s^2\,2s^2\,2p^6\,3s^1$

$3s^1$ orbitals has n = 3, $\ell = 0$

Number of radial nodes = $n - \ell - 1 = 3 - 0 - 1 = 2$

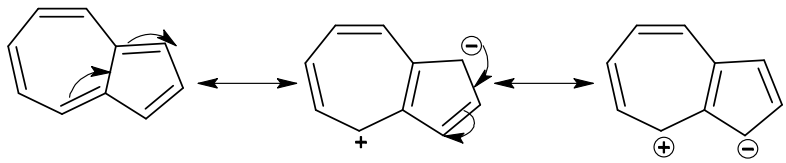
6. $$\frac{r_{O_2}}{r_{CH_4}} = \frac{P_{O_2}}{P_{CH_4}}\sqrt{\frac{M_{CH_4}}{M_{O_2}}} = \frac{n_{O_2}}{n_{CH_4}}\sqrt{\frac{16}{32}} = \frac{3}{32}\times\frac{16}{2}\sqrt{\frac{16}{32}} = \frac{3}{4\sqrt{2}}$$

7. Half life $t_{1/2}$ = 140 days

Number of half lives in total time = $\frac{280+140}{140} = 3$

$R = R_0(1/2)^n$

$\Rightarrow 3000 = R_0(1/2)^3$, R_0 = 24000 dps

9. 

11. $$n_{Cl_2} = \frac{5\times 8}{0.082\times 300} = 1.63$$

Mass of $Al_{(s)} = 0.05\times 10^{-1}\times(20\times 20\times 6)\times 1.13 = 13.56$ gm

So mass of $Al_{(s)}$ dissociated = 0.5 moles

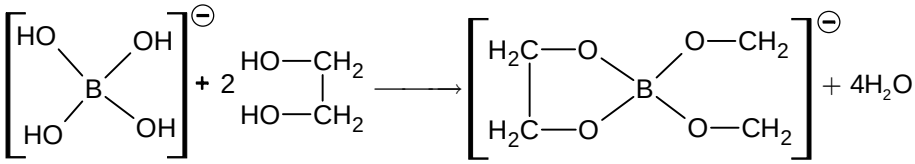
So eq is established

$$Al_{(s)} + \frac{3}{2}Cl_{2(g)} \rightleftharpoons AlCl_{3(s)}$$

$$1.63 - \frac{3}{2}\times 0.5 \quad 0.5$$

$$K_p = \left(P_{Cl_2}\right)^{-\frac{3}{2}} = \left(\frac{0.88\times 0.082\times 300}{8}\right)^{-\frac{3}{2}} = (2.7)^{-\frac{3}{2}}$$

(Volume of the vessel is taken 8 lit approx even after eq is established).

13. 

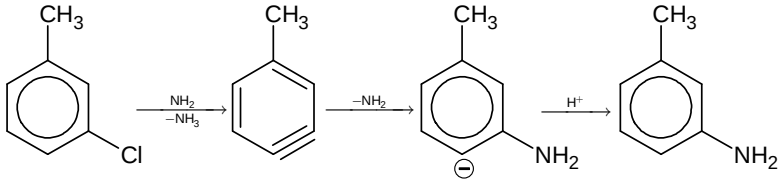
14. (i) $Na_2S_2O_3 + 2HCl \longrightarrow 2NaH + SO_2 + H_2O + S$

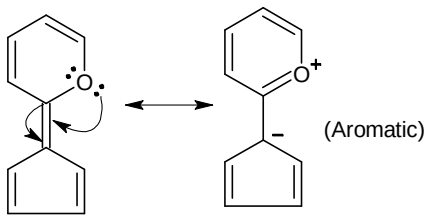
(ii) $KI_3 + 2Na_2S_2O_3 \longrightarrow KI + 2NaI + Na_2S_4O_6$

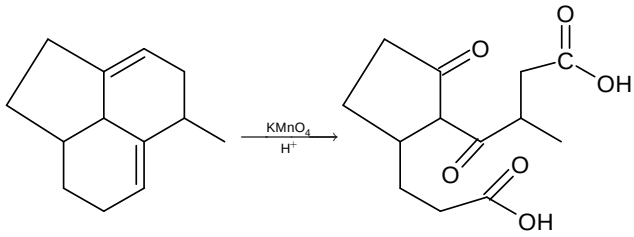
(iii) $2AgNO_3 + Na_2S_2O_3 \longrightarrow \underset{(white)}{Ag_2S_2O_3} + 2NaNO_3$

$$\xrightarrow{H_2O} \underset{Black}{Ag_2S} + H_2SO_4$$

15. Stability of intermediate carbocation increases rate of reaction.

16. 

18. 

19. 

21. $Mg_2C_3 + 2H_2O \longrightarrow Mg(OH)_2 + CH_3 - C \equiv CH$

24. $pOH = pK_b + \log_{10}\frac{[NH_4Cl]}{[NH_4OH]} = 4.7 + \log\frac{0.1}{0.01} = 5.7$

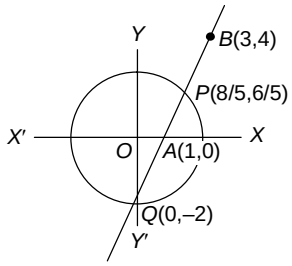
pH = 14 – 5.7 = 8.3

Mathematics

PART – III

SECTION – A

1. The equation of the line joining A(1, 0) and B(3, 4) is $y = 2x - 2$. This cuts the circle $x^2 + y^2 = 4$ at $Q(0, -2)$ and $P\left(\frac{8}{5}, \frac{6}{5}\right)$.



We have $BQ = 3\sqrt{5}, QA = \sqrt{5}, BP = \frac{7}{\sqrt{5}}$ and $PA = \frac{3}{\sqrt{5}}$

$\therefore \alpha = \frac{BP}{PA} = \frac{7/\sqrt{5}}{3/\sqrt{5}} = \frac{7}{3}$ and $\beta = \frac{BQ}{QA} = \frac{3\sqrt{5}}{-\sqrt{5}} = -3$

$\therefore \alpha, \beta$ are roots of the equation $x^2 - x(\alpha + \beta) + \alpha\beta = 0$

i.e., $x^2 - x\left(\frac{7}{3} - 3\right) + \frac{7}{3}(-3) = 0$ or $3x^2 + 2x - 21 = 0$.

2. $ax^2 + by^2 + 2fx + 2gy + c = 0$

$$a\left\{x^2 + \frac{2fx}{a}\right\} + b\left\{y^2 + \frac{2gy}{b}\right\} + c = 0$$

$$\Rightarrow a\left(x + \frac{f}{a}\right)^2 + b\left(y + \frac{g}{b}\right)^2 = \left(\frac{f^2}{a} + \frac{g^2}{b} - c\right)$$

$$\Rightarrow \frac{(x + f/a)^2}{\left(\frac{f^2}{a} + \frac{g^2}{b} - c\right)/a} + \frac{(y + g/b)^2}{\left(\frac{f^2}{a} + \frac{g^2}{b} - c\right)/b} = 1$$

i.e. e is eccentricity, then $\frac{\left(\frac{f^2}{a} + \frac{g^2}{b} - c\right)}{b} = \frac{\left(\frac{f^2}{a} + \frac{g^2}{b} - c\right)}{a}(1 - e^2)$

$$\Rightarrow 1 - e^2 = \frac{a}{b} \Rightarrow e^2 = \frac{b - a}{b} \therefore e = \sqrt{\left(\frac{b - a}{b}\right)}$$

3. $$\frac{(1-3x)^{1/2} + (1-x)^{5/3}}{2\left[1 - \frac{x}{4}\right]^{1/2}}$$

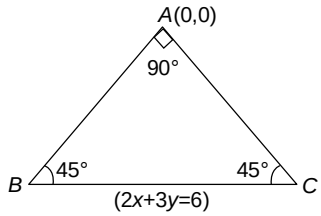
$$= \frac{\left[1 + \frac{1}{2}(-3x) + \frac{1}{2}\left(-\frac{1}{2}\right)\frac{1}{2}(-3x)^2 + \ldots.\right] + \left[1 + \frac{5}{3}(-x) + \frac{5}{3}\frac{2}{3}\frac{1}{2}(-x)^2 + \ldots.\right]}{2\left[1 + \frac{1}{2}\left(-\frac{x}{4}\right) + \frac{1}{2}\left(-\frac{1}{2}\right)\frac{1}{2}\left(-\frac{x}{4}\right)^2 + \ldots.\right]}$$

$$= \frac{\left[1 - \frac{19}{12}x + \frac{53}{144}x^2 - \ldots.\right]}{\left[1 - \frac{x}{2} - \frac{1}{8}x^2 - \ldots.\right]} = 1 - \frac{35}{24}x + \ldots.$$

Neglecting higher powers of x, then

$a + bx = 1 - \frac{35}{24}x \Rightarrow a = 1, b = -\frac{35}{24}$.

4. $y = mx$. It makes an angle of $\pm 45^o$ with $2x + 3y = 6$.



$$\therefore \quad \tan(\pm 45^o) = \frac{m-(-2/3)}{1+m(-2/3)} = \pm 1$$

or $\quad 3m + 2 = \pm(3 - 2m) \Rightarrow m = \frac{1}{5}, -5$

Hence sides are

$x - 5y = 0,$

$5x + y = 0$

and $2x + 3y = 6$.

Solving in pairs, vertices are $(0,0)$,

$$\left(\frac{6}{13}, \frac{30}{13}\right), \left(\frac{30}{13}, -\frac{6}{13}\right).$$

5. $L_{12} \equiv x - 3y + 1 = 0$

$L_{23} \equiv 2x + y - 12 = 0$

$L_{13} \equiv 3x - 2y - 4 = 0$

Therefore, the required distances are

$$D_3 = \left|\frac{4 - 3 \times 4 + 1}{\sqrt{10}}\right| = \frac{7}{\sqrt{10}}$$

$$D_1 = \left|\frac{4 + 1 - 12}{\sqrt{5}}\right| = \frac{7}{\sqrt{5}}$$

$$D_2 = \left|\frac{3 \times 5 - 2 \times 2 - 4}{\sqrt{9+4}}\right| = \frac{7}{\sqrt{13}} .$$

6. The equation of any line passing through the given point P(3, 4) and making an angle $\frac{\pi}{6}$ with x-axis is $\frac{x-3}{\cos 30^o} = \frac{y-4}{\sin 30^o} = r$ (say) .....(i)

Where 'r' represents the distance of any point Q on this line from the given point P (3, 4).

The coordinates (x, y) of any point Q on line (i) are

$(3 + r\cos 30^o, 4 + r\sin 30^o)$ i.e., $\left(3 + \frac{r\sqrt{3}}{2}, 4 + \frac{r}{2}\right)$

If the point lies on the line $12x + 5y + 10 = 0$, then

$$12\left(3 + \frac{r\sqrt{3}}{2}\right) + 5\left(4 + \frac{r}{2}\right) + 10 = 0 \Rightarrow r = \frac{132}{12\sqrt{3} + 5} .$$

7. Suppose a point on circle is $B(x_1, y_1)$ and that which divides A and B in 3 : 2 is P given by

$$h = \frac{-2 + 3x_1}{5}, k = \frac{2 + 3y_1}{5} \text{ or } \frac{5h + 2}{3} = x_1, \frac{5k - 2}{3} = y_1$$

As (x_1, y_1) lies on circle $x^2 + y^2 = 4$, we get on substituting, $25(x^2 + y^2) + 20(x - y) - 28 = 0$.

8. $(y - 3)^2 = -4 (x - 3)$ (given equation of parabola)

$\therefore$ equation of latus rectum is $x = 2$.

9. $x = t^2 + t + 1 \text{and} y = t^2 - t + 1$

$\Rightarrow x + y - 2 = 2t^2 \text{and} x - y = 2t$

$\Rightarrow 2(x + y - 2) = (x - y)^2 \Rightarrow x^2 + y^2 - 2xy - 2x - 2y + 4 = 0$

Comparing with the equation

$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0,$

$\therefore abc + 2fgh - af^2 - bg^2 - ch^2 \neq 0$ and $h^2 = ab$

10. Since the number of students giving wrong answers to at least i question $(i = 1, 2, ..., n) = 2^{n-i}$.

The number of students answering exactly $i (1 \le i \le -1)$ questions wrongly = {the number of students answering at least i questions wrongly, $i = 1, 2, ..., n)$} – {the number of students answering at least $(i + 1)$ questions wrongly $(2 \le i + 1 \le n)$} $= 2^{n-i} - 2^{n-(i+1)} (1 \le i \le n - 1)$.

Now, the number of students answering all the n questions wrongly $= 2^{n-n} = 2^0$.

Thus the total number of wrong answers

$= 1(2^{n-1} - 2^{n-2} + 2(2^{n-2} - 2^{n-3}) + 3(2^{n-3} - 2^{n-4})$

11. We have,

$y + z = a(\cos^2 x + \sin^2 x) + c(\sin^2 x + \cos^2 x) = a + c$

$(\therefore$ solution is (b)$)$

$y - z = a(\cos^2 x - \sin^2 x) + 4b \sin x \cos x - c(\cos^2 x - \sin^2 x)$

$= (a - c)\cos 2x + 2b \sin 2x$

$= (a - c).\left(\frac{1 - \tan^2 x}{1 + \tan^2 x}\right) + 2b.\left(\frac{2 \tan x}{1 + \tan^2 x}\right)$

$= (a - c).\left\{\frac{1 - 4b^2 / (a - c)^2}{1 + 4b^2 / (a - c)^2}\right\} + 2b.\left\{\frac{2.2b / (a - c)}{1 + 4b^2 / (a - c)^2}\right\}$

Since $\tan x = \frac{2b}{(a - c)}$,

$\therefore y - z = \frac{(a - c).\{(a - c)^2 - 4b^2\} + 8b^2(a - c)}{(a - c)^2 + 4b^2}$

$= \frac{(a - c)(a - c)^2 + 4b^2}{\{(a - c)^2 + 4b^2\}} = (a - c)$

$\Rightarrow y \neq z, (\because a \neq c)$.

12. Given that $z^2 + (p + iq)z + r + is = 0$ .....(i)

Let $z = \alpha$ (where α is real) be a root of (i), then

$\alpha^2 + (p + iq)\alpha + r + is = 0$

or $\alpha^2 + p\alpha + r + i(q\alpha + s) = 0$

Equating real and imaginary parts, we have $\alpha^2 + p\alpha + r = 0$ and $q\alpha + s = 0$

Eliminating α, we get $\left(\frac{-s}{q}\right)^2 + p\left(\frac{-s}{q}\right) + r = 0$

or $s^2 - pqs + q^2 r = 0$ or $pqs = s^2 + q^2 r$

13. $x = 3\left(\frac{1-t^2}{1+t^2}\right), y = \frac{2t}{1+t^2}$

$\frac{x^2}{9} + y^2 = \frac{\left(1-t^2\right)^2 + 4t^2}{\left(1+t^2\right)^2} = 1$ Which is an ellipse

$\therefore 1 = 9\left(1-e^2\right) \Rightarrow e^2 = 1 - \frac{1}{9} = \frac{8}{9}$

14. Here a = 5, b = 4 and e = $\frac{3}{5}$

Hence foci of the ellipse are ($\pm$ 3, 0)
Hence $PF_1 + PF_2 = 10$.

15. The given ellipse is $\frac{x^2}{4} + \frac{y^2}{3} = 1 \Rightarrow a = 2, b = \sqrt{3}$

$\therefore\ 3 = 4\left(1-e^2\right) \Rightarrow e = \frac{1}{2}$

$\therefore\ ae = 2 \times \frac{1}{2} = 1$

Hence, the eccentricity e_1, of the hyperbola is given by

$1 = e_1 \sin\theta \Rightarrow e_1 = \operatorname{cosec}\theta\ \Rightarrow\ b^2 = \sin^2\theta\left(\operatorname{cosec}^2\theta - 1\right) = \cos^2\theta$

Hence, equation of hyperbola is $\frac{x^2}{\sin^2\theta} - \frac{y^2}{\cos^2\theta} = 1$ or $x^2\operatorname{cosec}^2\theta - y^2\sec^2\theta = 1$

16. Since $(n+2)^{th}$ term is the middle term in the expansion of $(1+x)^{2n+2}$, therefore $p = {}^{2n+2}C_{n+1}$.
Since $(n+1)^{th}$ and $(n+2)^{th}$ terms are middle terms in the expansion of $(1+x)^{2n+1}$, therefore $q = {}^{2n+1}C_n$ and $r = {}^{2n+1}C_{n+1}$ But ${}^{2n+1}C_n + {}^{2n+1}C_{n+1} = {}^{2n+2}C_{n+1}$
$\therefore\ q + r = p$

17. For the first set number of ways ${}^{52}C_{17}$. Now out of 35 cards left 17 cards can be put for second in ${}^{35}C_{17}$ ways similarly for 3rd in ${}^{18}C_{17}$. One card for the last set can be put in only one way. Therefore the required number of ways for the proper distribution

$$= \frac{52!}{35!17!} \times \frac{35!}{18!17!} \times \frac{18!}{17!1!} \times 1! = \frac{52!}{(17!)^3}.$$

18. $\tan(\cot x) = \cot(\tan x)\ \Rightarrow\ \tan(\cot x) = \tan\left(\frac{\pi}{2} - \tan x\right)$

$(70)^2 + 20h + h^2 = (6)(70)(20)\ \ \cot x = n\pi + \frac{\pi}{2} - \tan x \Rightarrow \cot x + \tan x = n\pi + \frac{\pi}{2}$

$(70)^2 + 20h + h^2 = (6)(70)(20)\ \ \frac{2}{\sin 2x} = n\pi + \frac{\pi}{2} \Rightarrow \sin 2x = \frac{2}{n\pi + \frac{\pi}{2}} = \frac{4}{(2n+1)\pi}.$

19. $1 + \sin x + \sin^2 x + \ldots\infty = 4 + 2\sqrt{3}$

$$\Rightarrow \frac{1}{1-\sin x} = 4+2\sqrt{3} \Rightarrow \sin x = 1-\frac{1}{4+2\sqrt{3}}$$

$$\Rightarrow \sin x = 1-\frac{(4-2\sqrt{3})}{4} = \frac{2\sqrt{3}}{4} = \frac{\sqrt{3}}{2}$$

$$\Rightarrow x = \frac{\pi}{3} \text{ or } \frac{2\pi}{3}.$$

20. Adding and subtracting the given relation,

we get $(m+n) = a\cos^3\alpha + 3a\cos\alpha\sin^2\alpha + 3a\cos^2\alpha.\sin\alpha + a\sin^3\alpha$

$= a(\cos\alpha+\sin\alpha)^3$

and similarly $(m-n) = a(\cos\alpha-\sin\alpha)^3$

Thus, $(m+n)^{2/3} + (m-n)^{2/3}$

$= a^{2/3}\{(\cos\alpha+\sin\alpha)^2 + (\cos\alpha-\sin\alpha)^2\}$

$= a^{2/3}\{2(\cos^2\alpha+\sin^2\alpha)\} = 2a^{2/3}$.

21. We have $\sin A = n\sin B \Rightarrow \frac{n}{1} = \frac{\sin A}{\sin B}$

$$\Rightarrow \frac{n-1}{n+1} = \frac{\sin A-\sin B}{\sin A+\sin B} = \frac{2\cos\frac{A+B}{2}\sin\frac{A-B}{2}}{2\sin\frac{A+B}{2}\cos\frac{A-B}{2}}$$

$$= \tan\frac{A-B}{2}\cot\frac{A+B}{2}$$

$$\Rightarrow \frac{n-1}{n+1}\tan\left(\frac{A+B}{2}\right) = \tan\frac{A-B}{2}.$$

22. We have equal roots, therefore $\lambda^2 = 4\mu$.

Now second equation $x^2+\lambda x-12=0$ has a root 2, so put $x=2 \Rightarrow 4+2\lambda-12=0 \Rightarrow \lambda=4$

Hence from $\lambda^2 = 4\mu$, we have $\mu = \frac{16}{4} = 4$

$\Rightarrow (\lambda,\mu) = (4,4)$.

23. Let the roots are α and $n\alpha$

Sum of roots, $\alpha+n\alpha = -\frac{b}{a} \Rightarrow \alpha = -\frac{b}{a(n+1)}$ .....(i)

and product, $\alpha.n\alpha = \frac{c}{a} \Rightarrow \alpha^2 = \frac{c}{na}$ ....(ii)

From (i) and (ii), we get

$$\Rightarrow \left[-\frac{b}{a(n+1)}\right]^2 = \frac{c}{na} \Rightarrow \frac{b^2}{a^2(n+1)^2} = \frac{c}{na}$$

$\Rightarrow nb^2 = ac(n+1)^2$.

Note: Students should remember this question as a fact.

24. We have $\sqrt{3}+i = (a+ib)(c+id)$

$\therefore ac - bd = \sqrt{3}$ and $ad + bc = 1$

Now $\tan^{-1}\left(\frac{b}{a}\right) + \tan^{-1}\left(\frac{d}{c}\right)$

$= \tan^{-1}\left(\frac{\frac{b}{a}+\frac{d}{c}}{1-\frac{b}{a}.\frac{d}{c}}\right) = \tan^{-1}\left(\frac{bc+ad}{ac-bd}\right) = \tan^{-1}\left(\frac{1}{\sqrt{3}}\right) = n\pi + \frac{\pi}{6}, n \in I$

25. We know that $\log_a m > \log_a n \Rightarrow m > n$ or $m < n$, according as $a > 1$ or $0 < a < 1$. Hence for $z = x + iy$

$\log_{(1/3)} |z+1| > \log_{(1/3)} |z-1| \Rightarrow |z+1| < |z-1| \quad \left\{\because 0 < \frac{1}{3} < 1\right\}$

$\Rightarrow |x + iy + 1| < |x + iy - 1|$

$\Rightarrow (x+1)^2 + y^2 < (x-1)^2 + y^2$

$\Rightarrow 4x < 0 \Rightarrow x < 0 \Rightarrow \text{Re}(z) < 0$

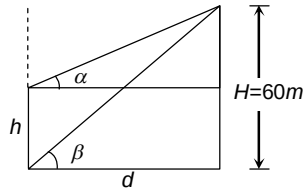
26. $x^{2n-1} + y^{2n-1}$ is always contain equal odd power. So it is always divisible by $x + y$.

27. Check through option, the condition $\left(\frac{n+1}{2}\right)^n \geq n!$ is true for $n \geq 1$.

28. $H = d\tan\beta$ and $H - h = d\tan\alpha$

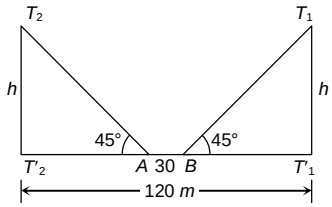
$\Rightarrow \frac{60}{60-h} = \frac{\tan\beta}{\tan\alpha} \Rightarrow -h = \frac{60\tan\alpha - 60\tan\beta}{\tan\beta}$

$\Rightarrow h = \frac{60\sin(\beta-\alpha)}{\cos\alpha\cos\beta\frac{\sin\beta}{\cos\beta}} \Rightarrow x = \cos\alpha\sin\beta.$



29. $\tan 45^o = 1 = \frac{h}{T_2'A} \Rightarrow T_2'A = h$

Hence $120 = h + 30 + h \Rightarrow h = 45$ m.



30. $x = \sqrt{1+\sqrt{1+\sqrt{1+.....}}}$ to ∞ ∞

We have $x = \sqrt{1+x}$

$\Rightarrow x^2 = 1 + x \Rightarrow x^2 - x - 1 = 0$

$\Rightarrow x = \frac{1 \pm \sqrt{1+4}}{2} = \frac{1 \pm \sqrt{5}}{2}$

As $x > 0$, we get $x = \frac{1+\sqrt{5}}{2}$